

CEE PRO v2 Whitepaper

A method note describing the volatility contraction, expansion breakout, and exhaustion-exit model used in the improved Pine Script v6 strategy.

RESEARCH AND DESIGN NOTE

Prepared April 24, 2026 | Pine Script v6 | Strategy template

Abstract

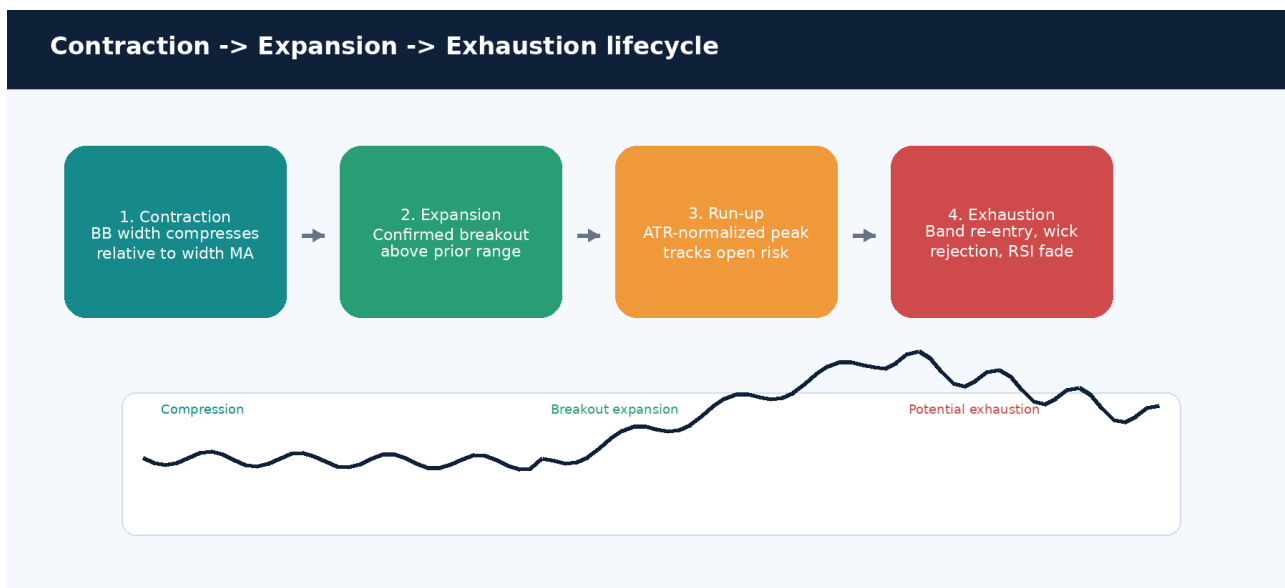
This whitepaper describes CEE PRO v2, a long-only TradingView strategy template based on a volatility lifecycle: contraction, expansion, run-up, and exhaustion. The design objective is not to predict every trend, but to create a transparent, testable process for identifying compression breakouts and managing late-stage breakout risk with ATR-normalized rules.

The model improves the original script by adding a scoring layer, optional market-quality filters, explicit state management, breakeven migration, time-based exits, richer alert payloads, and a dashboard for diagnostic monitoring.

Educational and research material only. The method requires independent validation before any use with real capital.

1. Research premise

Many liquid markets alternate between low-volatility balance and directional repricing. A contraction breakout strategy attempts to participate when volatility expands out of a compressed state. The primary risk is that not every expansion becomes a durable trend; some expansions end in exhaustion, rejection, or mean reversion. CEE PRO v2 therefore treats the exit engine as a first-class component, not an afterthought.



The design hypothesis

A higher-quality long setup should satisfy four conditions: volatility has compressed, price breaks a prior range, the expansion bar shows sufficient range and close strength, and the trade can be managed with predefined risk controls if momentum fails after run-up.

2. Signal definitions

The script uses normalized Bollinger Band width to detect contraction. Normalization divides band width by the Bollinger basis so that width is comparable across price levels. The strategy then compares that width to a moving average and to the lowest width over a lookback window.

Component	Definition	Purpose
EMA trend	Exponential moving average of close.	Directional filter and visual context.
BB width	(Upper band - lower band) / basis.	Volatility compression proxy.
ATR	Average True Range over the selected length.	Risk normalization for stops, run-up, and targets.
RSI	Relative Strength Index over the selected length.	Momentum roll confirmation for exhaustion.
Close location	(close - low) / (high - low).	Breakout strength and late-bar rejection proxy.
ADX / RVOL / VWAP / structure	Optional market-quality filters.	Reduce low-quality breakouts when empirically useful.

Contraction rule

```
bbWidth < bbWidthMa * contractionFactor
bbWidth <= lowest(bbWidth, contractionLookback)
```

Expansion rule

```
close > highest(high, breakoutLen)[1]
close > upperBand
barRange > ATR * expansionRangeATR
closeLocation >= minCloseLocation
```

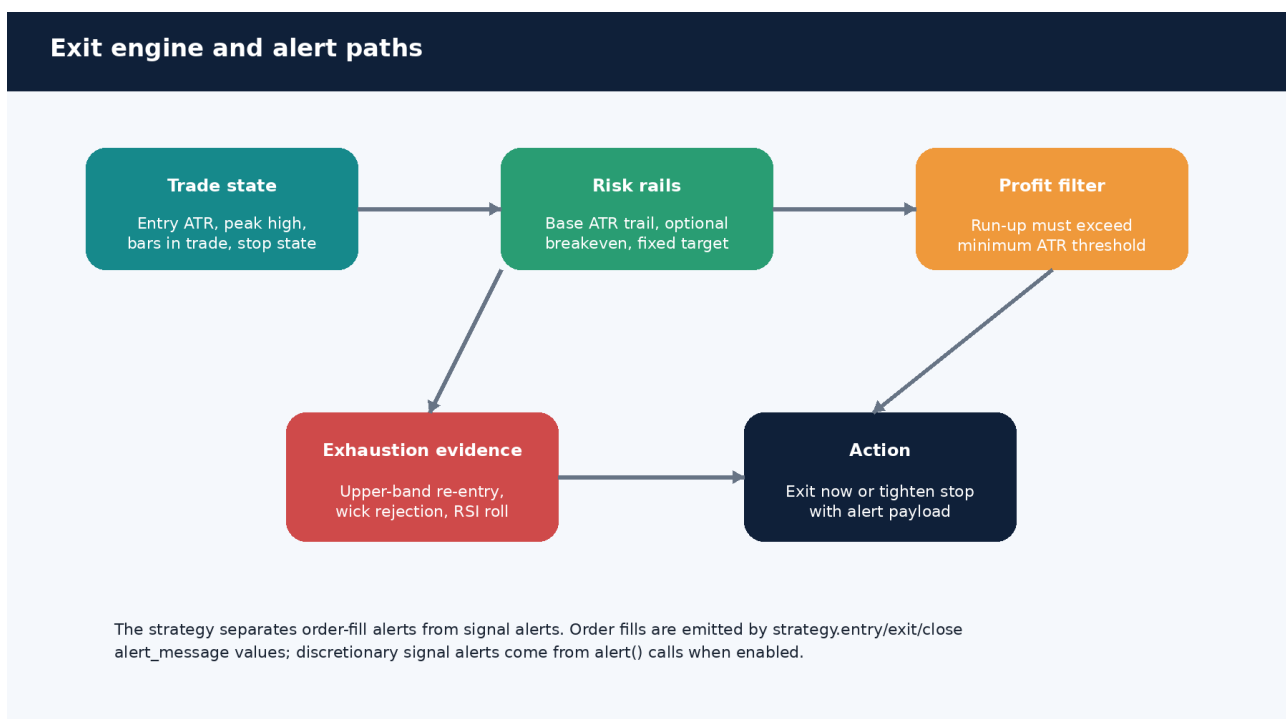
3. Scoring and filter architecture

The setup score is a diagnostic layer. It helps the user see why the system is waiting, armed, or eligible for entry. The final long entry still requires the core boolean gates to pass; the score is not a substitute for them.

Score feature	Weight	Interpretation
Recent contraction	15	Volatility compression occurred within the memory window.
Breakout above range	15	Price closed above the prior breakout lookback high.
Expansion bar	15	The breakout bar is wide enough and above trend context.
Close strength	15	The close is sufficiently high in the candle range.
Session / trend / volume / ADX / VWAP / structure	45 combined	Optional market-quality and context gates.

4. Trade state and risk engine

Once a long position opens, the strategy stores entry ATR, peak high, bars in trade, and whether the stop has been tightened. This explicit state allows the script to compute run-up and adjust the trailing stop without relying on discretionary chart reading.



Trailing stop

The base stop uses a chandelier-style ATR trail from the highest price observed after entry. When exhaustion is detected in Tighten stop only mode, the ATR multiple switches from `baseTrailATR` to `tightTrailATR`. Optional breakeven logic can move the stop toward entry after a smaller run-up threshold.

Fixed target and maximum-bars exit

A fixed ATR target is available for users who want a bounded reward objective. A maximum-bars exit is available for markets where stale breakout positions tend to decay. Both are optional because their usefulness varies widely by instrument and timeframe.

5. Exhaustion model

The exhaustion model activates only after the trade has achieved a minimum run-up in ATR units. This prevents the strategy from treating normal early noise as exhaustion. It then scores evidence that the expansion may be failing.

Evidence	Rule	Intuition
Band re-entry	high > upper band and close < upper band	Price expanded outside the band but failed to hold.
Upper-wick rejection	upper wick fraction exceeds threshold	Sellers absorbed or rejected the breakout high.
Momentum roll	RSI declines and candle weakens	Momentum is no longer confirming the move.
Close-location fade	close location falls below rejection threshold	The bar closes away from its high.

The threshold is deliberately multi-factor. A single wick or one weak close is often not enough; the script requires sufficient combined evidence and a profitable run-up before acting.

6. Alerting and automation design

CEE PRO v2 separates signal messaging from order-fill messaging. This is useful for manual traders and automation workflows. The order-fill path uses `alert_message` values attached to strategy order functions. The discretionary signal path uses `alert()` calls when enabled. Production automation should decide which path is executable and should ignore or separately route the other path.

Design concern	Mitigation in CEE PRO v2
Repainting concerns	Entry and exhaustion signals are gated by <code>barstate.isconfirmed</code> .
Ambiguous alerts	Messages identify entry, exit fill, exhaustion exit, tighten stop, or time exit.
Duplicate routing	<code>alert()</code> calls can be disabled if order-fill alerts are the only executable path.
Operational audit	Messages include score, close, stop, target, width, or RVOL values where relevant.

7. Validation methodology

The strategy should be validated as a hypothesis, not marketed as a finished system. The validation objective is to determine whether the edge survives unseen data, realistic costs, and different volatility regimes.

Professional validation workflow

	In-sample	Walk-forward	Forward test	Deployment
Metrics	Profit factor Expectancy Drawdown	Rolling windows Parameter drift Holdout data	Paper trades Live alert logs Slippage notes	Production caps Change log Kill switch
Robustness	Parameter grid Market regimes Trade count	Nested testing No retuning leak Stress spreads	Missed signals Webhook timing Data anomalies	Monthly review Retire rules Version control
Execution	Commissions Slippage Session rules	Gap behavior Stop fills Broker parity	Order routing Partial fills Latency	Size limits Broker errors Reconciliation
Governance	Hypothesis doc Risk budget Benchmark	Approval gates Review cadence Exception log	Incident notes Risk escalation Capital freeze	Audit trail User sign-off Archive

Recommended protocol

- 1 Define the market universe, timeframe, session, and cost assumptions before looking at optimization results.
- 2 Split historical data into in-sample and out-of-sample periods. Reserve a final holdout period if enough data exists.
- 3 Tune only a small number of logically related parameters at a time.
- 4 Use parameter grids and robustness plots rather than a single best setting.
- 5 Measure average trade, drawdown, profit factor, win/loss distribution, exposure, and trade count.
- 6 Forward-test alerts with paper execution and compare logs to chart markers before using live capital.

8. Limitations and risk considerations

- Compression can persist longer than expected; a breakout can fail quickly.
- Indicators such as Bollinger Bands, ATR, RSI, ADX, and VWAP are derived from historical price and volume; they do not know future liquidity or news.
- Backtests may not model spreads, rejected orders, partial fills, borrow constraints, margin changes, exchange outages, or broker-specific order behavior.
- Parameter optimization can create overfit results that look strong historically and degrade immediately in live use.
- Automation requires separate operational controls, including max position size, kill switches, broker reconciliation, duplicate-alert handling, and fail-safe logic.

9. Implementation summary

Module	Implementation summary
Setup detection	BB width contraction, range breakout, expansion bar, close-location filter.
Quality filters	Optional session, EMA slope, volume, ADX, VWAP, and confirmed swing-high gates.
State management	Tracks entry ATR, peak high, bars in trade, tightened-stop state, and exit cooldown.
Exit engine	ATR trail, optional breakeven, fixed target, maximum-bars exit, and exhaustion scoring.
Monitoring	Plots, labels, contraction background, dashboard table, and dynamic alerts.

10. References

- TradingView Pine Script v6 Language Reference Manual - <https://www.tradingview.com/pine-script-reference/v6/>
- TradingView Pine Script Strategies Concepts - <https://www.tradingview.com/pine-script-docs/concepts/strategies/>
- TradingView Pine Script Alerts Concepts - <https://www.tradingview.com/pine-script-docs/concepts/alerts/>
- TradingView Pine Script Bar States Concepts - <https://www.tradingview.com/pine-script-docs/concepts/bar-states/>
- TradingView Pine Script Tables Visuals - <https://www.tradingview.com/pine-script-docs/visuals/tables/>

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